FLRW-Departure Metrology for Bianchi Anisotropic Cosmologies: Exact Conventions, CAS-Verified Theorems, Statistical Architecture, Current-Data Constraints, and a Literature-Grounded Novelty Ledger

(External Research Report, v11)

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Abstract

This report is a self-contained scientific record of a programme that decomposes departures from a Friedmann–Lemaître–Robertson–Walker (FLRW) background into a typed comparator $x_C = \Sigma^2 - W^2 + \Omega_{\rm tilt} + \Delta\Omega_k$ of geometric, kinematic, tilt, and curvature contributions, and adjudicates each contribution through layers of exact geometry, transfer, identifiability, and calibrated inference. Relative to the v10 record it adds: (i) a constraint-natural vorticity convention and a frame-indexed component algebra, each certified by five independent computer-algebra kernels; (ii) a joint feasible-set support theorem and a species-resolved multi-fluid moment cone, likewise five-axis certified; (iii) a finite-window stochastic bulk-flow bridge whose identifiability rank is CAS-certified; and (iv) a novelty ledger in which every claim family is tiered K/C/P/S against a named study with an explicit delta. Data-lane constraints are reported at the completeness reached at the build-date probe; the DESI official-mock lane is stated as pending and no partial mock enters any statistic.

1 Scope and supersession

This document supersedes the v5–v10 external reports as the current self-contained record; those packages remain byte-frozen. It reports exact results (definitions and theorems with proofs), the statistical architecture, and current-data constraints at the completeness reached at the build-date probe. Two conventions and a family of theorems are certified by five independent computer-algebra systems—Wolfram with xAct, SymPy, SageMath with Singular, Lean 4 (kernel-checked `native_decide`), and Rocq/Coq (an independent kernel)—so that a symbolic result rests on two kernel-independent proof-assistant lineages, not on cross-checking one algebra engine against itself. No development-history, review-process, or promotion-status material appears here; provenance is limited to artifact hashes and literature citations.

2 Framework and frozen conventions

The background comparison uses the $1+3$ covariant decomposition. Writing Σ^2 for the (expansion-normalized) shear scalar, W^2 for the vorticity scalar, $\Omega_{\rm tilt}$ for the tilt energy, and $\Delta\Omega_k$ for the signed curvature departure, the generalized Friedmann/Gauss constraint rearranges to

$$x_C \;=\; \Sigma^2 - W^2 + \Omega_{\text{tilt}} + \Delta\Omega_k,$$

with the FLRW comparator at $x_C = 0$ and the sign vector $c = (1, -1, 1, 1)$ read off the constraint rather than postulated.

Proposition 1 (Constraint-natural vorticity convention). *With the dual vector* $\omega_a = \frac{1}{2}\varepsilon_{abc}\omega^{bc}$ *one has* $\omega_{ab}\omega^{ab} = 2\,\omega_a\omega^a$*, so under* $\Theta = 3H$ *the expansion-normalized vorticity scalar is* $W^2 := \omega_{ab}\omega^{ab}/(6H^2) = \omega_a\omega^a/(3H^2)$*, and* $\sqrt{\omega_{ab}\omega^{ab}}/\Theta \le B \Rightarrow W^2 \le \frac{3}{2}B^2$*.*

Proof. The dual identity gives $\omega_a\omega^a = \frac{1}{2}\omega_{ab}\omega^{ab}$, whence $\omega_{ab}\omega^{ab}/(6H^2) = \omega_a\omega^a/(3H^2)$. Squaring the hypothesis, $\omega_{ab}\omega^{ab} \le 9B^2H^2$, and dividing by $6H^2$ yields $W^2 \le \frac{3}{2}B^2$. The two forms and the ceiling are certified exactly by all five computer-algebra axes; the registered numerical anchor $W^2_{\max} = 3.3789222980376e-13$ is unchanged (a display convention, not a new value). □

The curvature contribution $\Delta\Omega_k$ is a *signed* scalar carrier; it is never projected onto a positive-semidefinite magnitude, which is reserved for the tensor 3-Ricci sector. Components are frame-indexed (n/matter/observer/electron), and a quantity in one frame is combined with a quantity in another only through a declared bridge with a validated remainder; a vortical congruence has no orthogonal hypersurface, so its rest-space carries the rest-bundle 3-Ricci rather than a hypersurface Gauss curvature.

3 Exact theorems

Proposition 2 (Joint feasible-set support). *For a compact common feasible set* F *the comparator identified set is the exact interval* $I_C = [\inf_F c^\top g,\ \sup_F c^\top g]$*; the per-axis product box is the corollary that holds only when* F *factorizes, and on a coupled* F *the joint interval is strictly narrower.*

Proof. $c^\top g$ is linear and F compact, so its image is the stated interval, attained at extreme points. If $F = \prod_j F_j$ the extremes separate and I_C equals the signed product box; otherwise a coupling constraint removes a box vertex from F, so $\sup_F c^\top g$ is attained on a lower-value face and the joint interval is strictly inside the box. On the exact rational fixture the five-axis CAS certifies the coupled joint $[0, 1]$ strictly inside the product box $[0, 2]$, with Lean and Rocq verifying the strict inequality over $\mathbb{Q}$. On the physical carrier the coupled manifold gives joint $x_C \in [0.08, 0.10]$ inside the marginal box $[0.04, 0.14]$. □

Proposition 3 (Species-resolved multi-fluid moment cone). *A vanishing first velocity moment (zero net flux) does not imply a vanishing second moment: two antipodal equal-weight streams have zero flux, positive trace (tilt energy), and a traceless nonzero anisotropic stress.*

Proof. For streams $\pm v\,e_x$ with unit weight the first moment is $v - v = 0$; the second moment is $K = \text{diag}(2v^2, 0, 0)$ with trace $2v^2 > 0$; the anisotropic stress $3\Pi = 3K - \text{tr}(K)\,I = \text{diag}(4, -2, -2)v^2$ is traceless $(4 - 2 - 2 = 0)$ yet nonzero. The same-trace isotropic comparator has zero stress, so the anisotropic stress, not the flux, distinguishes the two. Certified by all five CAS axes; Lean and Rocq verify the integer identities. □

Proposition 4 (Finite-window bulk-flow to homogeneous-tilt bridge). *The map from a homogeneous tilt to finite-window bulk-flow observables has rank* 3 *for a multi-window operator and rank* 1 *for a single window; hence one bulk-flow amplitude cannot point-identify the three-component homogeneous tilt, while a declared multi-window field bridge can.*

Proof. The window operator W has Gram matrix $G = (10W)^\top(10W)$ with $\det G = 394584 \neq 0$, so $\operatorname{rank} W = 3$; a single row is an outer product of rank 1. The generalized-least-squares estimator through the multi-window operator recovers the three-vector with small bias. The determinant and ranks are certified by all five CAS axes. □

Proposition 5 (MES attribution surface, frozen-anchored). *The Maartens–Ellis–Stoeger vorticity ceiling is a monotone attribution surface in the intrinsic rapidity ε_1; its intrinsic-zero endpoint reproduces the frozen geodesic ceiling $W^2_{\max} = 3.3789222980376e{-}13$ exactly, while the full-observed-dipole attribution is excluded by the hierarchy-preservation threshold. $\varepsilon_1 = 0$ is a source-backed branch choice, not a uniqueness theorem, and an unsourced coefficient is never a live ceiling.*

Proof. With $B_\omega(\varepsilon_1) = B_\omega(0) + \frac{10}{3}\varepsilon_1$ and $W^2 = \frac{3}{2}B^2_\omega$, anchoring $B_\omega(0) = \sqrt{2W^2_{\max}/3}$ makes $W^2(0) = W^2_{\max}$ exactly; the crossover $\varepsilon_{1,\mathrm{crit}} = \frac{43}{25}\varepsilon_2 + \frac{9}{35}\varepsilon_3$ sits below the full observed rapidity, so full-dipole attribution violates the hierarchy. The geodesic coefficients are the archived Stoeger–Araujo–Gebbie reduction; the print-only non-geodesic coefficients carry no accessible source and are quarantined. □

Proposition 6 (Rank-2 comparator non-identification and staged sharpness). *The declared response map to the four comparator components has rank 2 with a kernel spanned by the $(W^2, \Delta\Omega_k)$ directions, so those two channels are jointly unobservable pre-solver; and physical sharpness is proven in stages (algebraic → constraint → local), with the global Einstein–matter stage a registered obligation requiring a native solver.*

Proof. Two independent symbolic engines agree the response Jacobian has rank 2 with the stated kernel; family labels are exactly the quotient classes of the declared response, refined to singletons only by adding transverse-curvature and vorticity observables (rank ladder $2 \to 3 \to 4$). The sharpness ladder attains the algebraic (PSD witness) and constraint (exact zero momentum-constraint residual on homogeneous data) stages and registers the global stage as blocked, with stage-skipping refused. □

4 Statistical architecture

Proposition 7 (Partial-identification coverage). *For a set-identified scalar with a boundary of half-width w, the Imbens–Manski interval—widening by the constant C solving $\Phi(C + \Delta/\sigma) - \Phi(-C) = 1 - \alpha$—restores coverage of the least-favorable boundary at the nominal level, whereas a point-estimate Gaussian interval centered on the set midpoint undercovers as w grows.*

Proof. The least-favorable point sits at a boundary; the Imbens–Manski constant interpolates between the one-sided $z_{1-\alpha}$ and two-sided $z_{1-\alpha/2}$ critical values as Δ/σ ranges over $[0, \infty)$, giving nominal boundary coverage for every w. A Monte-Carlo grid confirms the IM interval holds ≥ 0.94 across regimes while the midpoint interval falls to zero coverage at large w. □

Proposition 8 (Cluster-exchangeable finite-null rank and anytime evidence). *When observations are exchangeable only within clusters, a valid finite-sample p ranks the observation among null draws that share the cluster structure and includes the observation, giving $p = (1 + b)/(N + 1)$; and a merged sequential e-value has expectation ≤ 1, so by Ville's inequality a running product crosses $1/\alpha$ with probability $\leq \alpha$ under any stopping rule.*

Proof. The within-cluster exchangeable null makes the observation's rank uniform on the support grid, so the conservative $(1 + b)/(N + 1)$ estimator never returns zero and never drops below the

$1/(N+1)$ resolution floor; a naive reuse of cluster labels as i.i.d. is refused. For the e-value, each factor is a likelihood ratio with unit expectation under the null; the product is a nonnegative martingale, and Ville's maximal inequality gives the crossing bound. Monte-Carlo confirms unit mean, Markov tail bounds, and the crossing bound. □

Proposition 9 (Estimated-covariance calibration and coherent evidence)**.** *A χ^2 formed with an inverse sample covariance is inflated by the Hartlap factor and is calibrated once corrected; and an inactive normalized nuisance parameter leaves the Bayesian evidence unchanged, so it can create no Occam penalty.*

Proof. The inverse Wishart expectation gives the $(N_{\rm sim} - m - 2)/(N_{\rm sim} - 1)$ correction, verified in simulation (raw mean $/m > 1.08$; corrected ≈ 1). A normalized prior integrates to one, so the marginal likelihood is invariant under adding it; an unnormalized prior shifts the log-evidence by its log-mass, and a duplicated response yields a zero log Bayes factor, precluding a manufactured geometry preference. □

Proposition 10 (Local/global source discrimination with mandatory abstention)**.** *Among explicit kinematic, large-scale-structure, survey-systematic, shared-global, and superposition competitors, a Bayesian-information-penalized model comparison with a look-elsewhere gap and a goodness-of-fit adequacy gate must abstain when the favored model is inadequate or the discrimination is degenerate; a confusable source is never assigned to the global competitor.*

Proof. Each competitor is a fixed linear model with a conjugate-Gaussian evidence; the penalized score selects a model only when the look-elsewhere margin is exceeded and the residual passes the adequacy gate, else it abstains. On a synthetic battery the superposition is recovered as the full model and the confusable-weak source abstains at rate one. □

5 Current-data constraints

Constraints are reported at the completeness reached at the build-date probe.

Planck low-multipole isotropy (K1). Under the real Planck PR3 FFP10 end-to-end null, the look-elsewhere global p over the six registered low-multipole statistics is 0.039 on the exchangeable support grid, above the 0.001 resolution floor—consistent with the isotropic end-to-end ensemble. The temperature BiPoSH diagonal vanishes identically for every odd L (an exchange-symmetry structural zero, verified by an independent Wigner-$3j$ oracle), so only even-L orientation terms carry trials; the pooled even-L diagonal log-power rank is $p = 0.195$, consistent with the end-to-end null.

ACT DR6 convergence isotropy. With the mean field debiased against the released simulation ensemble, the low-multipole ($\ell = 2$–10) debiased band power is consistent with the isotropic simulations ($p \approx 0.35$); the raw quadratic-estimator inference is a documented no-go because the release does not ship the raw filtered maps, so only an upper-limit/consistency constraint is reported.

CF4 peculiar-velocity flow. Under a same-data ΛCDM linear cosmic-variance covariance ($A^{-1}MA^{-1}$ with the fiducial velocity power spectrum), the estimator-matched cosmic variance dominates the measurement noise, deflating the naive flow significance to order one standard deviation; the depth-resolved flow is reported as an identified set that widens with depth and contains the isotropic point, and the growth difference across depth is not identified (reported as a bound, never as a precision tension). The two peculiar-velocity primary findings remain open pending independent non-author adjudication.

DESI number-count dipole. The window-corrected number-count dipole is reported only through the survey-conditional pooled-rank null (consistent), because the official validation-mock ensemble is still being acquired; causal attribution is deferred. No partial mock enters any rank, p-value, covariance, or significance.

6 Novelty ledger

Each claim family is tiered against the external literature: **K** known/textbook, **C** a cross-check of a specific named study, **P** a potential delta over existing research, **S** a significant delta. The programme is dominated by faithful cross-checks and re-derivations of named results with exact, machine-verified certificates; the tiers were assigned by an external-literature pass and are reported here as scientific provenance. The significance column ranks importance to the field and is independent of novelty.

Family	Result	Tier	Significance	Anchor
`T-XC`	Master FLRW-departure comparator identity	C	incremental	`EllisVanElst1999`
`T-W2`	Constraint-natural vorticity convention	K	incremental	`EllisMaartens201`
`T-EGS`	One-way FLRW/EGS and converse counterexamples	C	incremental	`Nilsson1999Almos`
`T-MES`	MES source-exact ceilings and attribution surface	C	incremental	`StoegerAG1997`
`T-OMK`	Ω_k higher-order slaving	P	incremental	`WainwrightEllis1`
`T-JOINT`	Joint feasible-set support theorem	K	marginal	`ImbensManski2004`
`T-RANK2`	Rank-2 comparator non-identification	K	marginal	`EllisMacCallum19`
`T-MULTIFLUID`	Multi-fluid moment cone	K	marginal	`IsraelStewart197`
`T-BRIDGE`	Finite-window bulk-flow bridge	K	marginal	`Watkins2009`
`T-PARITY`	Solver-free parity and handedness	C	marginal	`PontzenChallinor`
`T-FRAME`	Congruence-indexed frame algebra	K	incremental	`VanElstUggla1997`
`T-BIANCHI`	Bianchi class A/B atlas	K	marginal	`EllisMacCallum19`
`T-KE`	King–Ellis rotating congruence	P	incremental	`KingEllis1973`
`T-SHARP`	Physical sharpness ladder	K	marginal	`ChoquetBruhat195`
`M-PARTIALID`	Estimated-covariance partial identification	K	incremental	`ImbensManski2004`
`M-CLUSTER`	Cluster-exchangeable finite-null rank	K	incremental	`PhipsonSmyth2010`
`M-EVALUE`	Anytime e-value / Ville crossing	K	marginal	`Ramdas2023`
`M-ESTCOV`	Finite-covariance calibration	K	incremental	`Hartlap2007`
`M-EVIDENCE`	Coherent evidence and inactive-prior invariance	K	marginal	`MacKay2003`
`M-DISCRIM`	Source discrimination with abstention	C	incremental	`Secrest2021`
`D-K1`	Planck K1 low-multipole isotropy	C	incremental	`Planck2018VII`
`D-CF4`	CF4 flow and cosmic-variance deflation	C	incremental	`Watkins2023`
`D-DESI`	DESI number-count dipole	C	incremental	`Secrest2021`
`D-ACT`	ACT DR6 convergence isotropy	C	marginal	`Madhavacheril202`
`D-TEFF`	T$_\rm eff$ certified surrogate	K	marginal	`SpurioMancini202`

No family is tiered **S**: the load-bearing identities are textbook covariant kinematics, and the new content consists of exact machine-verified certificates and honest identifiability/coverage statements around named studies.

7 Analyses conditional on data completion

The following analyses are fully specified and run without design changes when their stated condition is met; none has been started on partial data.

1. **DESI official-mock causal attribution**: the exact-selection dipole machinery runs the pre-registered clustering/kinematic/selection attribution with per-mock refits once the complete official ensemble (1000 EZmock + 25 AbacusSummit) is authenticated (transfer at 390/1000 EZmock at the build probe; days-scale remaining).

2. **Independent non-author adjudication** of the peculiar-velocity estimator-level findings whose numerical instantiations are withheld above; they return with validated uncertainties or stay retired.

3. **Second-pipeline low-multipole replication** of the K1 result under an independent component-separation pipeline (bandwidth-bounded acquisition).

4. **Native anisotropic-transfer stage**: all family-identification, geometry, and handedness tests remain dormant until a native solver line delivers authenticated transfer functions; no diagnostic here substitutes for that stage.

8 Reproducibility

Every number in this report is loaded from a content-addressed result card. The report is regenerated by `scripts/build_external_audit_report_v11.py` (`--write`), with a byte-stable `--check` mode for the text artifacts (the PDF is excluded because the LaTeX toolchain embeds timestamps). Symbolic results carry a five-axis computer-algebra certificate under one contract hash; Lean and Rocq provide two kernel-independent proof-assistant lineages. The manifest ships the SHA-256 of every input card. Load-bearing inputs:

```
docs/generated/pr150_e2e_pooled_rank.json
docs/generated/k1_evenl_biposh_rank_card.json
docs/generated/act_kappa_card.json
docs/generated/pr186_result_card.json
docs/generated/pr187_result_card.json
docs/generated/pr189_result_card.json
docs/generated/pr197_result_card.json
docs/generated/pr200_result_card.json
docs/generated/pr217_result_card.json
docs/generated/pr222_result_card.json
docs/generated/pr223_result_card.json
docs/generated/claim_adjudication/literature_verdicts.json
```

The v5–v10 packages remain byte-frozen and are superseded by this record.